Skills Summary

Three Views of a Limit

Use this reference while completing your worksheet or when studying.

$\lim_{x \to a} f(x) = L$ means: as x gets close to a from both sides — *without ever equalling a* — the outputs get close to L. The value $f(a)$ is irrelevant. It exists exactly when $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$. Tables give evidence, graphs give a picture, algebra gives proof.

1. From a table of values

Sample inputs closing in on a from *both* sides. If both columns settle on the same number, that is what the limit should be — but a table sees only finitely many inputs, so it can never prove one.

Example: A table for $f(x) = \dfrac{x^2 - 25}{x - 5}$ gives $f(4.99) = 9.99$ and $f(5.01) = 10.01$. Find $\lim_{x \to 5} f(x)$.

Step 1. Both one-sided columns close on the same value, so read the trend instead of evaluating at the missing point.

Step 2. The entries straddle 10 and tighten toward it; cancelling $\dfrac{(x-5)(x+5)}{x-5} = x + 5$ confirms it. $\Rightarrow$ **10**

Try it: A table for $f(x) = \dfrac{x^2 - 49}{x - 7}$ gives $f(6.99) = 13.99$ and $f(7.01) = 14.01$. Find $\lim_{x \to 7} f(x)$.

check: **14**

2. From algebra

Substitute first: a number *is* the limit. $\frac{0}{0}$ means a hidden common factor — rewrite, then substitute. Factor and cancel ($a^2 - b^2 = (a-b)(a+b)$, $a^3 - b^3 = (a-b)(a^2+ab+b^2)$); use the conjugate when a root causes it; divide by the highest power when $x \to \infty$. Cancelling changes the function at one point, never the limit.

Example: $\lim_{x \to 3} \dfrac{x^2 - x - 6}{x - 3}$

Step 1. Substitution gives $\frac{0}{0}$, a signal to rewrite rather than an answer, so factor and hunt for the common factor.

Step 2. $\dfrac{(x-3)(x+2)}{x-3} = x + 2$ for $x \neq 3$, and at $x = 3$ that is $3 + 2$. $\Rightarrow$ **5**

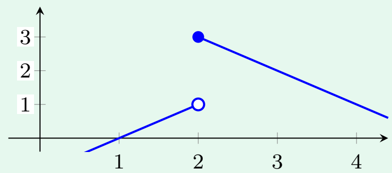
Try it: $\lim_{x \to 3} \dfrac{x^3 - 27}{x - 3}$

check: **27**

3. From a graph

Trace the curve toward $x = a$ once from each side and read the *height* you head for, not the dot. An open circle means the point is missing (the limit can still exist); a jump between the sides means the two-sided limit does not exist.

Example: From the graph of q, find $\lim_{x\to 2^-} q(x)$.



Step 1. The minus sign asks only about inputs left of 2, so follow the left branch and ignore the dot on the right.
Step 2. That branch is the line $y = x - 1$, climbing to the open circle at height 1. $\Rightarrow$ **1**

Try it: From the same graph, find $\lim_{x\to 2^+} q(x)$. Do the two sides agree?

check: **3** — no, so the two-sided limit does not exist

4. A series sum is a limit

An infinite sum is *defined* as the limit of its partial sums: $\sum_{n=0}^{\infty} a_n = \lim_{k\to\infty} S_k$ with $S_k = a_0 + \cdots + a_k$. Geometric ($a_n = ar^n$) has $S_k = a\frac{1-r^{k+1}}{1-r}$, so for $|r| < 1$ the r^{k+1} term dies and $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$; for $|r| \geq 1$ the limit fails and the series diverges.

Example: Evaluate $\displaystyle\sum_{n=0}^{\infty} 5\left(\tfrac{1}{3}\right)^n$.

Step 1. Each term is the previous one times a fixed ratio, so it is geometric and the partial-sum limit has a closed form.
Step 2. $a = 5$, $r = \frac{1}{3}$, $|r| < 1$, so $S = \dfrac{5}{1 - \frac{1}{3}} = \dfrac{5}{\frac{2}{3}}$. $\Rightarrow$ $\boldsymbol{\frac{15}{2}}$

Try it: Evaluate $\displaystyle\sum_{n=0}^{\infty} 4\left(\tfrac{1}{5}\right)^n$.

check: **5**